Microsoft Fabric PMO & EVM Analytics

End-to-End Operational & Project Controlling Guide | Lakehouse • Direct Lake • PBIP Reporting

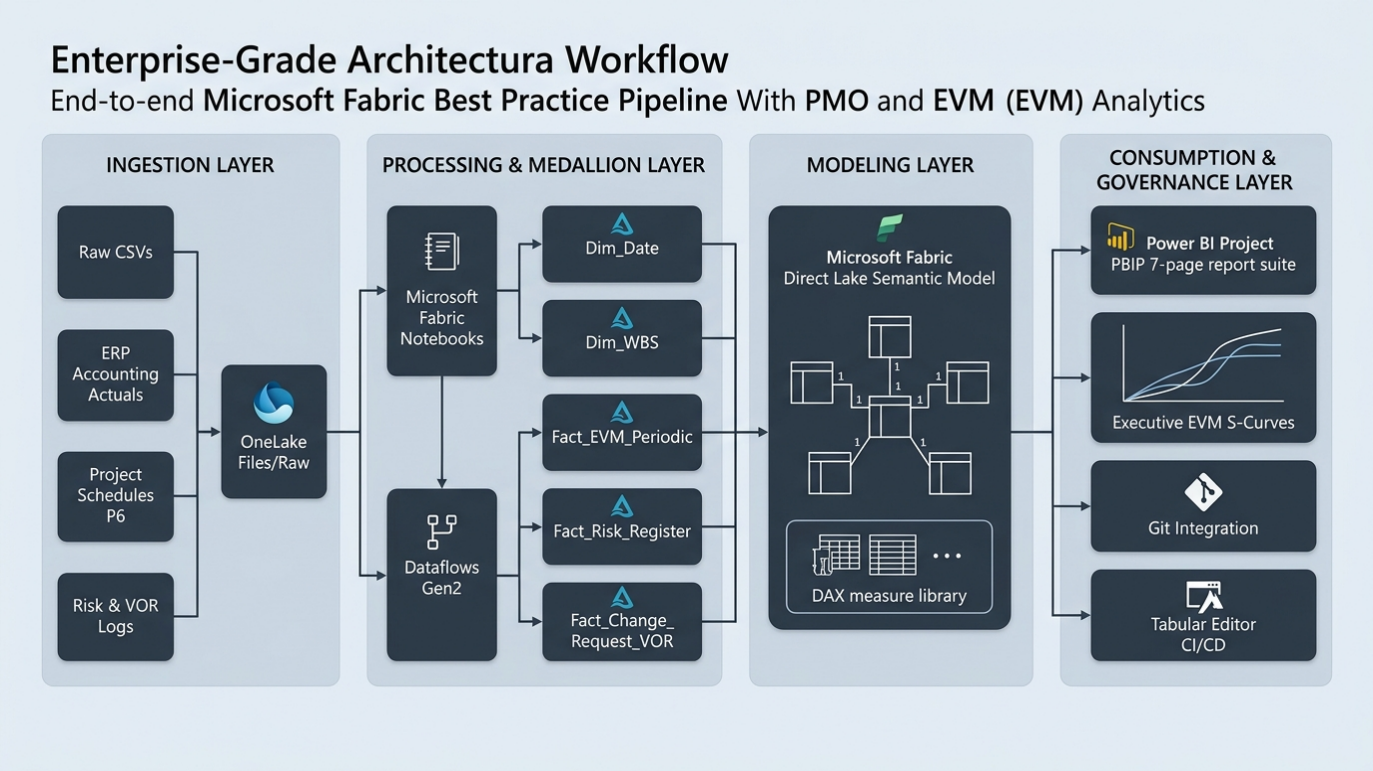
Workspace: PMO_EVM_Analytics | **Lakehouse:** lh_pmo_evm | **Semantic Model:** PMO_EVM_Model

Project Domain: Skjold-Class Stealth Fast Patrol MTB (Defense / Maritime) | **Total BAC:** 766.00 M NOK (24 Months)

Lead Controller / Workspace Admin: Frank Ellingsen | **Version:** 1.2 | **Date:** August 2026

1. Microsoft Fabric Architecture & Data Pipeline

This project establishes a standardized, enterprise-grade analytical pipeline in Microsoft Fabric for Earned Value Management (EVM) and integrated project risk/scope governance. It connects disparate project management systems (ERP accounting actuals, Primavera P6 schedules, risk registers, and change order databases) through a high-performance Medallion Delta Lake architecture.



Pipeline Stage Breakdown:

- **1. Ingestion Layer (OneLake Files / Raw):** Automated landing of raw CSVs and transactional extracts without schema locking.
- **2. Medallion Processing Layer (Delta Lake):** Cleansing, deduplication, incremental merging, and ACID-compliant Delta table creation via Fabric PySpark Notebooks and Dataflows Gen2.
- **3. Modeling Layer (Direct Lake):** Zero-latency semantic model querying Delta parquet files in OneLake with a Star Schema (1:N single-directional) and 44 central DAX measures.
- **4. Consumption Layer (Power BI PBIP & Web):** 7-page Tufte-compliant Power BI report suite with native Git Integration and CI/CD via Tabular Editor 3.

2. Lakehouse Structure & Data Schema

The Lakehouse (lh_pmo_evm) serves as the centralized storage and compute engine in Microsoft Fabric. All data is stored in open Delta Lake / Parquet format with V-Order optimization.

Table Name	Type	Granularity	Key Columns	Description / Business Purpose
Dim_Date	Dimension	Monthly (24 mo)	Date_Key, Date	Calendar dimension spanning project lifecycle (Jan 2025 – Dec 2026).
Dim_WBS	Dimension	Work Package (19)	Task_ID, WBS_Code	Work Breakdown Structure (L1/L2), Task Name, CAM, and Baseline Budget (BAC).
Fact_EVM_Periodic	Fact	WP x Month (456)	Fact_ID, Task_ID, Date_Key	Monthly periodic Planned Value (PV), Earned Value (EV), and Actual Cost (AC).
Fact_Risk_Register	Fact	Risk Item (7)	Risk_ID, Task_ID	Quantitative risk register with probability, expected cost impact, and EMV.
Fact_Change_Request_VOR	Fact	Change Order (8)	VOR_ID, Task_ID	Variation Order Requests, claimed vs approved costs, and schedule delay extensions.

Direct Lake Star Schema Relationships

- **Fact_EVM_Periodic[Date_Key]** → **Dim_Date[Date_Key]** (Many-to-One, Single)
- **Fact_EVM_Periodic[Task_ID]** → **Dim_WBS[Task_ID]** (Many-to-One, Single)
- **Dim_WBS[Task_ID]** → **Fact_Risk_Register[Task_ID]** (One-to-Many, Bi-directional cross-filter)
- **Dim_WBS[Task_ID]** → **Fact_Change_Request_VOR[Task_ID]** (One-to-Many, Bi-directional cross-filter)

3. EVM Mathematical Formulations & DAX Library

All calculations are centralized in the _Measures TMDL table. Formulations strictly follow defense/aerospace standards (ANSI/EIA-748).

Metric	DAX Formulation / Logic	Project Controller Interpretation
PV_Cum	CALCULATE([PV], FILTER(ALL(Dim_Date), Dim_Date[Date_Key] <= MAX(Dim_Date[Date_Key])))	Baseline budget scheduled to date.
EV_Cum	IF(Date <= CutoffDate, CALCULATE([EV], ALL(Dim_Date) <= CurrentDate), BLANK())	Budgeted cost of work completed to date.
AC_Cum	IF(Date <= CutoffDate, CALCULATE([AC], ALL(Dim_Date) <= CurrentDate), BLANK())	Actual expenditures incurred to date.
CV	[EV_Cum] - [AC_Cum]	Cost Variance: Underspend (>0) vs Overspend (<0).
SV	[EV_Cum] - [PV_Cum]	Schedule Variance in currency units: Ahead (>0) vs Lagging (<0).
CPI	DIVIDE([EV_Cum], [AC_Cum], 1.0)	Cost Performance Index: Incurred cost efficiency factor.
SPI	DIVIDE([EV_Cum], [PV_Cum], 1.0)	Schedule Performance Index: Earned velocity efficiency factor.
EAC (CPI)	DIVIDE([BAC], [CPI])	Estimate at Completion assuming current cost efficiency continues.
ETC	MAX(0, [EAC] - [AC_Cum])	Estimate to Complete: Projected remaining cash requirements.
VAC	[BAC] - [EAC]	Variance at Completion: Expected total project variance.
TCPI (BAC)	DIVIDE([BAC] - [EV_Cum], [BAC] - [AC_Cum])	Required cost efficiency on remaining work to finish on budget.
Active EMV	CALCULATE(SUM(EMV), Status IN {'Active', 'Open'})	Quantified expected monetary loss from open project risks.

4. Power BI Project (PBIP) Report Suite Overview

The Power BI reporting solution uses the Enhanced Report Format (PBIR) split across 7 dedicated views. Each view targets specific governance levels, from steering committee overviews to Control Account Manager (CAM) ledgers.

#	Report Page	Primary Audience	Key Metrics & Visual Capabilities
1	**Executive EVM**	Steering Comm. / Leadership	12-Card Scorecard, EVM Triple S-Curve (PV/EV/AC/EAC), CPI/SPI Trends, Variance Drivers Table, Governance Watchlist.
2	**S-Curve & Burn Rate**	Project Controller / PMO	Full Direct-Labeled S-Curve, Monthly Incurred Spend vs Earned Velocity, Net Cost Variance Delta Waterfall.
3	**Schedule & Gantt**	Planning / Project Controls	WBS Milestone Progress, Plan vs Actual Dates, Planned % vs Earned %, SPI by Work Package, Delay Threat Days.
4	**WBS Swimlane**	Engineering CAMs	Discipline Lifecycle Matrix (Level 1 & 2), Cost Performance Index by Discipline, Budget vs Spend Distribution.
5	**Risk Matrix**	Risk Manager / PMO	Probability vs Cost Impact Heatmap, Active EMV Exposure by Category, Risk Lifecycle Status, Mitigation Action Ledger.
6	**VOR Governance**	Commercial / Contracts	Variation Order Request Ledger, Claimed vs Approved Cost Impacts, Approval Volume Breakdown, Schedule Extensions.
7	**Master WBS Register**	Project Accounting / Audit	Complete 16-Column PMO Accounting Ledger for all 19 WBS elements with full EVM formulas and RAG indicators.

Edward Tufte Data-Ink Visualization Standards

- **Zero Vertical Gridlines:** Tables and matrices remove vertical lines to speed horizontal scanning.
- **Direct Ink Labeling:** S-Curves label lines directly at cutoff and project end instead of remote legends.
- **High-Contrast Numbers:** Right-align all numeric values with tabular/monospace digits.
- **Restrained Palette:** Muted neutrals for baselines; Red/Amber reserved exclusively for negative variances (> ±5%).

5. Monthly Project Controller Operational Cadence

To maintain audit compliance, data integrity, and reliable forecasting, follow this monthly 5-step closing cadence:

Step 1: Accounting Cutoff & Actuals Validation (Day +2)

- Extract general ledger accounting actuals (AC) from ERP and upload to Lakehouse Files/Raw/Fact_EVM_Periodic.csv.
  - Verify total cumulative actual cost against financial accounting ledgers (ensure zero unposted accruals).

Step 2: CAM Progress Updates & Earned Value Assessment (Day +4)

- Collect physical progress percentages from Control Account Managers (CAMs).
  - Compute Earned Value (EV = BAC x % Complete) and update periodic increments in Lakehouse.

Step 3: Risk & Variation Order (VOR) Reconciliation (Day +6)

- Review monthly Risk Register updates: probability shifts, mitigated threats, and revised cost/schedule impacts.
  - Log newly approved Variation Orders in Fact_Change_Request_VOR and adjust contractual baselines if needed.

Step 4: Statistical EAC & Forecasting Review (Day +8)

- Evaluate statistical EAC (BAC / CPI) against CAM bottom-up estimates.
  - If CPI < 0.95 or SPI < 0.95, require a formal Variance Analysis Report (VAR) from the CAM with corrective action plans.

Step 5: Executive Package & Steering Committee Publishing (Day +10)

- Sync Fabric Lakehouse Direct Lake model.
  - Export executive PDF and PowerPoint briefing packs for management and steering committees.

6. Version Control, Git Integration & Tabular Editor

Microsoft Fabric native Git Integration enables enterprise software lifecycle management (ALM) for BI assets.

Git Best Practices for PBIP & Fabric

- **Never commit session cache:** Ensure .gitignore excludes **/.pbi/localSettings.json and **/.pbi/cache.abf.
- **Feature Branching:** Develop new measures or report layouts in feature branches (e.g. feature/cpi-waterfall).
- **Tabular Editor Automation:** Use scripts in scripts/ for batch DAX formatting, lineage tagging, and metadata validation.
- **Fabric Workspace Sync:** Connect Fabric Workspace directly to GitHub/Azure DevOps main branch for automated deployment.

Repository Directory Mapping:

Folder Path	Description & Role in Pipeline
data/raw/	Raw CSV inputs (Dim_Date, Dim_WBS, Fact tables) uploaded to Lakehouse.
data/database/	Local SQLite transactional store (mtb.db) for rapid offline verification.
dax/	Central DAX scripts (all_measures_dax_query_view.dax, reference files).
docs/	Architectural documentation, layout specifications, ERD, and PDF guides.
powerbi/	PBIP project containing .Report (PBIR) and .SemanticModel (TMDL).
scripts/	Tabular Editor automation scripts (import_measures_tabular_editor.cs).
web/	Standalone interactive HTML EVM report (pmo_evm_report.html).